

Traian Florin Șerbănuță

Formal Modelling & Verification · Functional Programming · Type Systems

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Experience

Consultant and Researcher

2024–2026

Pi Squared, Inc.

Core Rust engineer on verifiable-computing and settlement infrastructure.

- **Blocks → Circom pipeline** — Rust toolchain compiling a DSL into ZK subcircuits with end-to-end certificates.
- **FastSet validator + proxy** — validator and client-facing proxy for the FastSet protocol and verifiable settlement layer.
- **Metamath proof checkers** across several zkVM backends.
- **Fast Shop (agentic AI commerce)** — Universal Commerce Protocol integration with an MCP-exposed commerce surface; built largely with **Claude Code**.

Consultant and Researcher

2012–2023

Runtime Verification, Inc.

- **K Framework** — a DSL and compiler for executable programming-language semantics.
- **K Haskell symbolic-execution prover** — reachability-logic prover with SMT integration, applied to the EVM, WebAssembly, and IELE semantics.
- **CBC-Casper in Coq** — one of the first machine-checked safety proofs for the protocol.

Associate Professor of Computer Science

2013–Present

University of Bucharest

Teach **Haskell-based declarative programming**, **programming-language semantics**, and **program verification**; designing a graduate **AI-assisted Systems Design** course for Fall 2026.

Co-founder/VP, Institute for Logic and Data Science.

Skills

- Functional programming and type systems
- Programming-language semantics
- Compilers and DSLs
- Formal methods and verification
- Interactive theorem proving (Coq/Rocq, Lean)
- AI-assisted software development

Programming Languages

- Haskell
- Rust
- Coq / Rocq
- Java
- C / C++
- Lean 4
- TypeScript

Education

PhD in Computer Science

2010

University of Illinois, Urbana-Champaign

Designed and prototyped the **K Framework** — rewriting-based programming-language semantics (advisor: Grigore Roșu).

Open-source and Publications

- **Coq / Lean formalizations** — aml-in-coq (matching logic), semantics-in-lean, propositions-as-types.
- **Publications** — 55 articles, h-index 22, 2106 citations (Google Scholar · DBLP, 2026).